

PROJECT PROPOSAL (FIFTH YEAR-IDP)

	Contents		
Title of research	Design and Implementation of a Host-Based WiFi Intrusion Detection System for Shared Network Environments		
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Approach (Max:300 words)	Objectives: 1. To develop a real-time monitoring system that can detect common network threats such as ARP Spoofing, Port Scanning and Evil Twin attacks by analyzing live traffic patterns on the device. 2. To create a dual-mode alert system that provides both stationary and portable security, using an external hardware alarm for physical warnings and a screen-based notification for use in public places like cafes or libraries. 3. To design a user-friendly dashboard that allows non-technical users to easily see their network health and security status without needing to have superior knowledge in network technology and security.		
	Background including brief literacy review: In shared network places such as internet cafés, libraries and small businesses, the main computers used by owners or administrators are often connected to public Wi-Fi. These computers handle important tasks and stay online for long periods, which makes them easy targets for cyber-attacks. Common threats include Man-in-the-Middle attacks like ARP Spoofing and unauthorized port scanning. In Myanmar, many small business owners do not aware of network security or find it too difficult to set up, leaving their systems exposed. To solve this problem, this project proposes a Host-Based Intrusion Detection System. The system runs directly on the computer, so it does not require router access or advanced network permissions. It monitors network traffic in real time and analyzes packets to detect suspicious activities such as ARP Spoofing, Evil Twin attacks and active scanning, even on open or unsecured networks. The main feature of this project is its two-level response. The software		

	displays on-screen alerts so the owner can understand the attack, and it also connects to an external hardware module that triggers a physical alarm sound. This alarm helps warn and discourage attackers by showing that their actions are being monitored. Overall, this project provides a simple and affordable security solution for Windows-based systems in shared networks.
	Materials to be applied: 1. Router 2. External Hardware components for alarm system 3. Personal Computers with Linux and Windows Operating System 4. USB Wifi Adapter
Uniqueness (in case)	Wifi Intrusion Detection System (WIDS) with Hardware-Based Attacker Deterrence
Abstract Including Purpose (Max: 100 words)	This project proposes a WiFi Monitoring and Alarm System aimed at protecting wireless networks from network attacks and performance disruptions. The system is designed specifically for non-technical users, providing real-time detection of intrusions and cyber-criminal activities. Users can easily monitor, manage, and secure their WiFi networks without advanced technical knowledge. The purpose of this project is to empower non-technical users to secure their WiFi networks effectively. It aims to simplify complex networking concepts into an easy-to-use alarm and monitoring system that alerts users to potential threats, improves network reliability, and enables better administrative control.
5 Keywords about the project	Network Security, Window Software, Attack Detection Alarm System, Surveillance System
Expected results including Appealing points	The expected results are 1. Real-Time Attack Detection: The system continuously monitors the WiFi network and quickly detects unauthorized access, intrusion attempts, and suspicious activities, allowing users to respond before serious damage occurs. 2. User-Friendly Interface: The interface is designed for non-technical users, using simple language, icons, and alerts so that anyone can understand and manage WiFi security easily. 3. WiFi Performance Monitoring: The system tracks network performance and identifies interference or unusual traffic that may slow down the connection, helping maintain stable and reliable WiFi service. 4. Smart Alert System: The system provides clear and timely notifications through alarms or messages, informing users about threats and recommended actions in a simple and understandable way. In summary, the expected results of this project include a user-friendly

	WiFi security system for non-technical users. It provides real-time attack detection, improves WiFi performance by identifying interference, and strengthens security. Clear alerts and simple controls help users monitor and protect their networks confidently without technical expertise.
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